

WP-11a · Prior-art scout for the RPP / proof-of-understanding primitive

WP-11a (rpp-prior-art, supports WP-11 rpp-adversarial-paper) · tier internal ·
2026-07-18

The Privacy-is-Value Research Programme

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The RPP adversarial paper (WP-11) must not assert novelty it cannot defend. This scout ran in two passes. Pass 1 tested the five auth-primitive candidates the E3 extraction exposes (N1-N5, §1-§2) and returned all-MIRAGE: read as “comprehension-as-authentication,” RPP is crowded. The First Person ruled that framing wrong (2026-07-18): the auth-primitive axis is not where the novelty lives. Pass 2 (§0 below) re-scouted the corrected axes and found a genuinely stronger position. §0 governs; §1-§2 are retained as the evidence for what WP-11 must NOT pitch.

0. Corrected framing (First-Person ruling 2026-07-18) — this governs

RPP’s novelty is not “prove you understand” as an access test. It is three things the pass-1 framing never tested:

- **M1 — prompt injection as a constructive mechanism.** RPP embeds a directive in content that FORCES the reader/model to perform a specific cognitive act (author a relationship-proverb binding their context to the content) as a precondition of use — the injection mechanism turned into a cooperative feature. The point is NOT unforgeability: a model CAN perform the act, and that is fine; the novelty is compelling the cognitive act by injection-as-design. **Verdict: DEFENSIBLE and the cleanest leg.** The prompt-injection literature is ~99% threat-framed (Greshake et al. arXiv:2302.12173; Perez-Ribeiro arXiv:2211.09527; OWASP LLM01); the only constructive uses are ADVERSARIAL-DEFENSIVE — Tracebit “context bombs” (2026-07-14, the strongest neighbour, derails a *hostile* agent) and canary/tripwire honeypots (detect leakage) — or SELF-STEERING CoT (“Let’s think step by step,” Kojima et al. arXiv:2205.11916, which steers the operator’s *own* model, not a directive embedded in third-party content to gate a reader). No prior work uses cooperative injection to compel a legitimate reader to perform a designated generative/relational act as a use-precondition. **Mandatory carve-out:** the paper must distinguish RPP from context-bombs (constructive-but-adversarial) and from CoT (forcing-but-self-directed) explicitly; the gap is narrowing (2026 work), so pin the distinction precisely.
- **M2 — a proverbial surface over hard crypto. RULED: Option C, derive-from-the-pair (First Person, 2026-07-18, gate G-RPP-ARCH discharged).** The proverb is neither a single-narrative seed (Option A) nor a pure pointer over an independent key (Option B). It is **one half of a two-party derivation**: the trust object is the interleaved pair, $\kappa = \text{hash}(\text{interleave}(P_A,$

P_B)), and no single party ever holds the whole. A single proverb is a half; the counterparty is the second factor. This resolves the derive-vs-bind question by dissolving it — the construction *derives*, but from a secret that is *split across two parties*, so the brain-wallet entropy model (single low-entropy human input against a public target) does not apply: guessing one party's proverb yields nothing, and reconstruction requires both halves held by different people.

- **The interleaved visibility ratio (RULED into the design).** The two proverbs are combined by the **interleaved inscription path** (the third of the three modes: symmetric / asymmetric / interleaved — fragments interwoven across the visibility boundary), parameterized by a **disclosure ratio $\sigma \in [0,1]$** : the public anchor carries the σ -fraction of the interleaved pair, and the two parties jointly hold the $(1-\sigma)$ that stays private. σ becomes a tunable knob on the *shared knowledge* — trading verifiability/recoverability by the anchor against how much of the joint understanding stays between the two parties. This is disclosure-budget control on a bilateral knowledge-sharing artifact, which no located neighbour has. (Deployment status per E3-C32: the interleaved path + σ budget are DESIGN-layer; the deployed system currently uses the asymmetric default + stratum-derived visibility — WP-11 states this honestly. The ϕ sweet-spots for σ (0.382 / 0.5 / 0.618) are the register's conjectural C54 disclosure-ratio values, cited as candidate parameters, not asserted optima.)
- **Novelty under the ruling: CLEAN.** Against pure ENCODING layers (BIP-39, PGP word list, mnemonikey, what3words: bits→words, zero comprehension) M2 is safe. Against the decider — **TDCcommons defensive publication article 9969** (verified this session: DERIVES a key FROM a *single* user's narrative via NLP → embedding → LSH → fuzzy-extractor → deterministic key) — M2 is now clearly distinct: TDCcommons is single-narrative, passive-input, no disclosure knob; RPP is bilateral, injection-forced, and carries a tunable interleaved disclosure ratio. The pair-split is the security story; the interleaved σ is the knowledge-sharing control; neither is in TDCcommons.
- **M3 — relational-memory recovery material.** A proverb from a genuine shared experience is stored in and recalled through relational/episodic memory, materially more durable for human recall than random-word mnemonics (the axis on which BIP-39 seeds fail users). **Verdict: PREMISE IS TEXTBOOK, systems-synthesis UNRESOLVED.** The recall advantage is fully predicted by the generation effect (Slamecka-Graf 1978, DOI 10.1037/0278-7393.4.6.592 — the strongest neighbour; self-generated items are recalled better) and levels-of-processing (Craik-Lockhart 1972, DOI 10.1016/S0022-5371(72)80001-X), so M3 CANNOT claim the effect. It may claim only the *systems application* — relational/episodic material as crypto recovery material that beats random mnemonics by construction — and only after a SOUPS/USEC/USENIX-Security usable-crypto-backup sweep confirms no one has run that comparison (NOT yet run; A4/next-scout item). Carry M3 as supporting design rationale, not a headline novelty claim.

The distinguishers that survive in every case (from pass 1, now load-bearing): the trust object is the pair of two non-identical, independently-authored compressions, never compared for equality (N2 residue — Mayrhofer-Gellersen owns “independent derivation from shared context,” only the non-equality survives); and RPP forces the act by injection at point-of-use (M1). TDCcommons-9969 is single-narrative and takes its input passively; RPP is bilateral and compels the act. Even in the derive case, bilaterality + injection-forcing + relational memory distinguish RPP from every located neighbour — but that is a NARROW, honest claim, not “no prior work does comprehension-based recovery.”

1. The gap grid: trust-basis × verification-mode (pass-1 auth-primitive axis, superseded by §0)

Every access/credential scheme found sits somewhere on two axes. TRUST-BASIS = what the scheme rests the trust on. VERIFICATION-MODE = how it checks. RPP's claimed home is the bottom-right cell.

trust-basis verify-mode	retrieval / equality	proof-of-knowledge (extractor)	generative expansion vs. commitment
possession of a stored secret	passwords; SLIP-39 shares; social recovery (Argent, ERC-4337)	Schnorr, GMR ZK-PoK; PAKE (EKE, OPAQUE); Camenisch- Lysyanskaya credentials	(empty)
recall of a fact	KBA / security questions (Ellison, Just-Aspinall, Schechter); personal entropy	—	—
recognition / biometric / context	Déjà Vu; fuzzy commitment (Juels-Wattenberg); fuzzy extractors (Dodis-Reyzin-Smith); Shake-Well-Before- Use; secret handshakes	Weinshall cognitive auth	—
generic humanness (cognitive act)	CAPTCHA/reCAPTCHA; Naor Turing-test-auth; GOTCHA; Human Challenge Oracle (2026)	—	GOTCHA (labels vs. commitment)
comprehension of specific committed content	—	—	RPP (claimed); ICAE / LLMLingua-2 / Trukhina-Vashkelis 2026 occupy the mechanism; nobody occupies it AS a named auth credential

The surviving cell is not empty — it is contested. The {comprehension-of-specific-content × generative-expansion} cell is mechanically occupied by 2024-2026 LLM context-compression work (ICAE, gist tokens, LLMLingua-2) and, most sharply, by Trukhina-Vashkelis 2026 (“Compress the Context, Keep the Commitments”), which already pairs verifiable expansion with a “commitment.” What is NOT occupied: that mechanism reframed AS an authentication/credential primitive named

“proof of understanding,” bound to a per-principal commitment. RPP’s residue lives in the reframing, not the mechanism.

2. Novelty ledger — the five candidates, adjudicated

Verdict vocabulary (WP-14a house): VALIDATED = the stated claim survives as novel; MIRAGE = anticipated, but a narrower residue survives; BLOCKED = withheld. All five below are MIRAGE-with-residue. Ranked by adjudicated confidence that the residue itself survives, strongest first.

N4 (strongest residue) — MIRAGE · “prove comprehension, not possession, by generative expansion against a commitment”

Claim as stated. Verification is a generative rehydration/expansion of a compressed token against a public commitment, standing in for proof of knowledge of a secret, and unforgeable without genuine understanding. **Why it MIRAGE’d.** Two mechanical direct hits: GOTCHA (Blocki-Blum-Datta, AISec 2013) already binds a human-generated response to a stored commitment, explicitly unforgeable without a live human; and ICAE (Ge et al., ICLR 2024) already regenerates a source from a compressed token as the fidelity signal. The verifiable-commitment framing is itself taken (Trukhina-Vashkelis, arXiv:2605.17304, 2026). The word “**unforgeable**” is **dead**: expansion is reproducible by any equally-capable model (ICAE/gist/LLMLingua), and “unforgeable proof of knowledge” is owned by ZKP (GMR/Schnorr) for a stricter guarantee. **Surviving residue (narrow).** Binding a comprehension proof to a *specific required generative expansion, as an authentication/credential primitive named proof-of-understanding, keyed to a per-principal commitment* — no work in the context-compression cluster fuses the expand-a-compression mechanic with an auth credential under that framing. **Required rewrites:** drop or qualify “unforgeable” (use “capability-gated,” or add a genuine cryptographic binding); foreground the auth-primitive framing, not the mechanism. **Strongest threat:** GOTCHA (arXiv:1310.1137) decides the referee’s verdict; ICAE (arXiv:2307.06945) and Trukhina-Vashkelis (arXiv:2605.17304) narrow the residue.

N2 — MIRAGE · “the independent bilateral compression-pair as the credential”

Claim as stated. Two parties independently author a natural-language compression of a shared experience; the pair (not either alone, not a co-signature) is the hashed trust object, with independence-of-authorship as an anti-forwarding / anti-collusion property. **Why it MIRAGE’d.** “Two parties independently derive a token from a shared experience with no pre-shared secret” is published: Mayrhofer-Gellersen “Shake Well Before Use” (IEEE TMC 8(6), 2009) is a DIRECT HIT, with fuzzy extractors, context/co-presence pairing, and secret handshakes as strong NEARs. Every one of them, however, converges the two inputs onto the *same* token and verifies *equality* (or additive reconstruction). **Surviving residue (narrow).** The **non-equality** of the pair: RPP takes two deliberately divergent, independently-authored compressions, never compares them for equality, and makes the unreconciled pair itself the trust object, with independence weaponized as an explicit anti-forwarding property rather than a physics artifact (Shake-Well) or a co-signature/endorsement (proof-of-personhood vouching). State N2 exactly as: “the trust object is the pair of two non-identical, independently-authored compressions, never compared for equality, such that neither authoring alone and no co-signature substitutes.” Drop any claim to “independent derivation from shared context” broadly — Mayrhofer-Gellersen owns it. **Strongest threat:** Mayrhofer & Gellersen, IEEE TMC 2009.

N1 — MIRAGE · “comprehension as the access primitive replacing a stored secret”

Claim as stated. The access primitive is relocated from possession of a stored secret to demonstrated comprehension. **Why it MIRAGE’d.** The space is densely occupied and the category is already named: Naor’s Turing-test-auth (1996), CAPTCHA (von Ahn et al., 2003), Weinshall’s cognitive authentication (IEEE S&P 2006), the whole KBA line (Ellison et al. 2000; Just-Aspinall 2009; Schechter et al. 2009), and Bonneau et al. “The Quest to Replace Passwords” (IEEE S&P 2012) which enumerates “cognitive authentication” as an established category. “Knowledge not possession” is owned since Fiat-Shamir/Schnorr (1986-89). **Surviving residue (narrow).** Access keyed to demonstrated comprehension of a *per-principal public commitment with no recoverable stored witness* — not “authenticate by what you know,” not “prove humanness by a cognitive act.” This is a repackaging with a thin residue; it is the weakest standalone candidate and should be carried only as part of the conjunction, never as a headline. **Strongest threat:** Bonneau et al. 2012 (names the category); Weinshall 2006 (nearest mechanism).

N5 — MIRAGE · “recovery by regenerating a semantic token from remembered meaning”

Claim as stated. A lost credential is recovered by re-deriving a semantic token from remembered comprehension of a shared experience plus a public anchor, replacing seed-phrase/secret custody. **Why it MIRAGE’d.** The reproduce-from-noisy-input primitive is 20+ years old: fuzzy extractors (Dodis-Reyzin-Smith, EUROCRYPT 2004), fuzzy commitment (Juels-Wattenberg, CCS 1999), biometric key regeneration (Kanade et al., BTAS 2009). The seed-phrase-killing goal is crowded (Buterin social recovery; Argent; ERC-4337/MPC; SLIP-39) but those recover from externally-held shares, cleanly distinct from RPP’s cognitive input. **Surviving residue (narrow).** The *source model*: remembered comprehension/meaning of a shared experience as the noisy input to a regeneration scheme, framed as proof-of-understanding — no located precedent for that exact source. Novelty in the source and framing, layered on standard primitives. RPP must cite fuzzy extractors as substrate, not claim the primitive. **Strongest threat:** see the honesty obligation below (this residue is real but its SOUNDNESS is the problem, not its novelty).

N3 (weakest residue) — MIRAGE, near-BLOCKED · “compression ratio as a quantified comprehension metric”

Claim as stated. Compression ratio is a quantified, continuous measure of comprehension (“high compression = strong assessment”). **Why it MIRAGE’d, hard.** This is a named, decades-old thesis, not a new claim: Chaitin “comprehension is compression” (2006), Zenil “Compression is Comprehension” (2019, literally the title), institutionalized by the Hutter Prize (2006), formally grounded by Delétang et al. “Language Modeling Is Compression” (ICLR 2024), and *empirically validated as a continuous linear metric* by Huang et al. “Compression Represents Intelligence Linearly” (COLM 2024, Pearson ≈ -0.95), with 2025-2026 agent extensions. It is DOUBLE-EDGED: it pre-empts the general claim AND the semantic-compression critique (arXiv:2306.02305) contradicts it — compression optimizes predictive utility, not meaning, so “high compression = strong understanding” is contestable on its face. **Surviving residue (thinnest).** Only a re-scoping to a narrow instrument survives: compression ratio as a *per-agent assessment credential for a specific artifact*, explicitly NOT a claim about compression-and-comprehension in general — and the very paper it resembles (Huang et al.) validates the correlation at *model scale over corpora*, not per-response, so RPP cannot borrow that support for a single-artifact assessment. This candidate should be DEMOTED from a contribution to a design choice; asserting it as novel invites a referee to call it both unoriginal and overclaimed.

Strongest threat: Huang, Zhang, Shan, He, COLM 2024 (arXiv:2404.09937) — it IS N3, published and quantified.

3. The honesty obligations, RECLASSIFIED under the corrected framing (First-Person ruling)

Pass 1 surfaced three obligations as soundness liabilities. The ruling reclassifies all three: two dissolve into the corrected framing, one is conditional on the M2 architecture ruling. This is the point of the reframe — the auth-primitive framing manufactured liabilities the surface-over-crypto framing does not carry.

1. **The brain-wallet critique → DISCHARGED by the Option C ruling.** Pass 1 treated RPP as a credential whose security rests on one party's proverb entropy — the brain-wallet graveyard (Vasek-Bonneau, "The Bitcoin Brain Drain," FC 2016: ~98% drained). Option C removes the target the critique needs: the secret is the interleaved PAIR, split across two parties, and no single party holds it. Guessing one proverb yields nothing; the $(1-\sigma)$ private portion requires both halves held by different people. The brain-wallet threat model (a single low-entropy human input against a public target) simply does not have an object to attack. What replaces the liability is a DEFENSE WP-11 states positively: *unlike single-narrative brain wallets and unlike TDCCommons-9969, the recoverable secret is never held by, nor derivable from, one party — the interleaved pair binds two independent authorings, and the public anchor discloses only σ of it.* The two brain-wallet failure modes are each handled on their own axis: **memorability** by M3 (relational recall), **entropy** by the two-party split plus the $(1-\sigma)$ held fraction. The honest residual is now narrow and quantitative-in-A3, not a standing liability: WP-11's formalism must state security as a function of the split and $(1-\sigma)$, and confront the one real residual — that a co-participant of the shared experience holds one half by construction, so the threat model is a *malicious counterparty*, not an outside guesser (which is the correct, and stronger, framing).
2. **"AI can fake comprehension" → RECLASSIFIED as missing the point (dissolves).** Pass 1 treated "unforgeable without understanding" as the claim, then found it under fire from Personhood Credentials (Adler et al., arXiv:2408.07892) and Human Challenge Oracle (Maleki et al., arXiv:2601.03923). The ruling: unforgeability was never the claim. M1's mechanism is *forcing the cognitive act* by injection; that a model performs the act is expected and fine — the forcing is the feature. The word "unforgeable" is dropped (the pass-1 finding was right about the word, wrong about it being a liability). What remains is a real design note, not a liability: the forced act must be *bound to the specific content* so a generic completion does not satisfy it — this is the M1-vs-CoT carve-out, an engineering requirement, not a soundness hole.
3. **"Compression optimizes prediction, not meaning" → STANDS, but demoted with N3.** The semantic-compression limit (arXiv:2306.02305) still contradicts a general "high compression = strong understanding" claim. The ruling did not contest the N3 demotion: compression ratio is an internal instrument, not asserted as a general comprehension measure. This obligation is discharged by not making the claim.

Net: the corrected framing carries ONE conditional honesty item (the derive-case entropy argument, gated on the M2 ruling) instead of three standing liabilities. The reframe did not hide the liabilities; it showed two of them were artifacts of pitching RPP as an auth primitive rather than a proverbial surface over hard crypto.

4. Verdict — the defensible contribution under the corrected framing

WP-11's lead is M1, not a comprehension-credential. Stated at the altitude the pass-2 sweep defends:

A construction that uses **prompt injection as a cooperative mechanism** — a directive embedded in content that forces a reader/model to author a relationship-proverb binding their context to the content, as a precondition of use [M1] — where the trust object is the **interleaved pair of two non-identical, independently-authored proverbs** [Option C + N2], $\kappa = \text{hash}(\text{interleave}(P_A, P_B))$, split across two parties with a **tunable disclosure ratio σ** governing how much of the shared knowledge the public anchor carries versus what the two parties jointly hold [M2, interleaved-visibility]; the recovery material is **relationally/episodically memorable** rather than random [M3, design rationale]. The injection compels the act; the two-party split carries the security; the interleaved σ tunes the disclosure; the relationship carries the recall.

Novelty ranking, strongest first: **M1 (cleanest — cooperative injection is unoccupied)** > Option-C interleaved bilateral pair (N2 + the σ disclosure knob — the pair-split is the security story, distinct from every single-input neighbour) > M3 (systems-synthesis, pending the SOUPS/USEC sweep; premise is textbook). None is written as “no prior work does X.”

A5-in-PC-persona resilience. The claim survives the strongest citation on each axis — Tracebit context-bombs (M1), TDCommons-9969 (M2), Mayrhofer-Gellersen (N2), Slamecka-Graf (M3) — IF WP-11: (i) carves M1 out from context-bombs (constructive-but-adversarial) and CoT (forcing-but-self-directed) explicitly; (ii) frames the security as the Option-C two-party interleaved split — malicious-counterparty threat model, not outside-guesser — and states the disclosure/security relation as a function of σ and $(1-\sigma)$ in A3's formalism; (iii) frames M3 as design rationale until the usable-security sweep runs; (iv) drops “unforgeable” and does not pitch the comprehension-credential (§1-§2) as the contribution; (v) states the interleaved-path + σ -budget deployment status honestly (design-layer per E3-C32). With those moves the paper stands on M1 + the Option-C interleaved bilateral construction, which no located work occupies. The gate (G-RPP-ARCH) is now RULED, so move (ii) is a writing task, not a blocker.

This is a genuinely stronger position than pass 1 reported. The auth-primitive framing found only narrow MIRAGE residues; the corrected framing finds one clean novel leg (M1) that the first sweep never tested, because RPP was mis-cast as authentication when it is a constructive-injection + proverbial-surface mechanism. The lesson for the pipeline (feeds E10/WP-27): a prior-art sweep is only as good as the novelty framing it is handed — pass 1 was adversarially correct about the wrong axis. The A10 discipline held (it did not flatter), but the AXIS came from E3's auth-primitive claims; the First Person supplied the correct axis, and re-scouting it was mandatory before assertion.

4b. Outstanding verification (before WP-11 tier-A use)

- **The M2 architecture ruling (First Person): RULED 2026-07-18 — Option C, derive-from-the-interleaved-pair, with the tunable visibility ratio σ (gate G-RPP-ARCH discharged).** No longer blocking; it is now a writing task for A3 (state security as a function of the two-party split and $(1-\sigma)$) and a design note (the interleaved inscription path + σ budget are E3-C32 design-layer, deployment status stated honestly).
- **TDCommons-9969 full text (A4):** the PDF 403'd; the abstract/index confirms narrative→embedding→LSH→f extractor→deterministic-key (single-narrative derive). Obtain the full text via an authenticated fetch and read it before finalizing M2 — it is the decider citation, and Option C's

distinction from it (bilateral + interleaved σ) should be checked against the full claim set.

- **SOUPS/USEC/USENIX-Security sweep (next scout):** “memorable/story-based crypto key backup usability” — close M3 before any novelty claim on it.
- **The three 1970s memory DOIs (A4):** verify Slamecka-Graf 1978, Craik-Lockhart 1972, Rogers-Kuiper-Kirker 1977 at primary records.
- **The interleaved path + σ disclosure budget (A3/A1 cross-check):** the interleaved inscription mode is E3-C10; the $\sigma \in [0,1]$ disclosure budget is E3-C22 (design-layer, not deployed); the ϕ sweet-spots are register C54 (conjectural). WP-11 cites these at their stated status; a widening-sweep of E3 may promote the interleaved+ σ mechanism from design-note to a first-class extraction claim now that it is ruled load-bearing.

5. Must-engage prior art (the WP-11 related-work spine; A4 verifies before tier-A use)

- **Comprehension/cognitive-gate lineage:** Naor 1996 (Turing-test-auth); von Ahn et al. CAPTCHA (EUROCRYPT 2003) + reCAPTCHA (Science 2008); Weinshall cognitive auth (IEEE S&P 2006); Blocki-Blum-Datta GOTCHA (AISec 2013, arXiv:1310.1137).
- **Proof-of-knowledge / credentials:** GMR (SIAM J. Comput. 1989); Fiat-Shamir (CRYPTO 1986); Schnorr (CRYPTO 1989); Camenisch-Lysyanskaya (CRYPTO 2002).
- **Key-from-shared-context / bilateral:** Bellare-Meritt EKE (IEEE S&P 1992); Juels-Wattenberg fuzzy commitment (CCS 1999); Dodis-Reyzin-Smith fuzzy extractors (EUROCRYPT 2004); Mayrhofer-Gellersen Shake-Well (IEEE TMC 2009); Balfanz et al. secret handshakes (IEEE S&P 2003).
- **KBA / recovery:** Ellison et al. personal entropy (2000); Just-Aspinall (SOUPS 2009); Schechter et al. (IEEE S&P 2009); Bonneau et al. Quest to Replace Passwords (IEEE S&P 2012); Buterin social recovery (2021); SLIP-39.
- **Compression-as-comprehension:** Chaitin (2006); Zenil (2019); Hutter Prize (2006); Delé-tang et al. (ICLR 2024); Huang et al. (COLM 2024, arXiv:2404.09937); semantic-compression limit (arXiv:2306.02305).
- **LLM context-compression / expansion:** Ge et al. ICAE (ICLR 2024, arXiv:2307.06945); Mu-Li-Goodman gist tokens (NeurIPS 2023); Pan et al. LLM-Lingua-2 (ACL 2024); Trukhina-Vashkelis (arXiv:2605.17304, 2026).
- **AI-vs-cognitive-test / personhood:** Adler et al. Personhood Credentials (arXiv:2408.07892, 2024); Maleki et al. Human Challenge Oracle (arXiv:2601.03923, 2026); Borge et al. proof-of-personhood (IEEE EuroS&PW 2017).
- **The honesty bookend:** Vasek-Bonneau-Castellucci-Keith-Moore, The Bitcoin Brain Drain (FC 2016, DOI 10.1007/978-3-662-54970-4_36).

Provenance flag (D2): these citations were verified at primary records during the sweep but carry the litreview-corpus caveat — A4 re-verifies author/venue/year for each before it enters the tier-A WP-11 related-work section.